

AP CSP | Lesson 22: Efficiency & Project

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: C

C follows one-based indexing. A is the first value. B is the second. D is the fourth and could reflect zero-based indexing.

Q2. Answer: D

D is exponential growth. Constant, linear and square/polynomial growth are in the reasonable-time category. This classification concerns growth, not a guarantee of an instant run.

Q3. Answer: B

B states the practical trade-off. A removes the lack of optimality guarantee. C confuses optimization with decidability and overclaims. D invents a resource guarantee.

Q4. Answer: A

A states the universal limitation. B and C concern resources or missing implementation. D is too strong: some individual instances can be solved.

Q5. Answer: A and C

A is the binary-search prerequisite. C is parameterized reuse. B confuses sensing with movement. D confuses returning with displaying. Award one mark only for A and C together.

Q6. Answer: B

B counts 30 and 45. A excludes equality. C counts every item regardless of condition. D sums qualifying minutes instead of counting qualifying sessions.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: values[1]=10; LENGTH=4; visited values 10,20,30,40 in that order; four body executions.

1 mark: Gives values[1] as 10.

1 mark: Gives length 4.

1 mark: Gives traversal order 10,20,30,40.

1 mark: Gives four body executions.

Q8. 4 marks

Answer: (a) Sorted data. (b) 19 then 34. (c) Position is unchanged. (d) The robot remains in its current cell and the program terminates.

1 mark: States that binary-search data must be sorted.

1 mark: Gives inspected values 19 then 34.

1 mark: States that a left rotation does not change the cell.

1 mark: States both remaining in place and program termination after an invalid forward move.

Q9. 4 marks

Answer: n^2 gives 64 operations; 2^n gives 256. Polynomial n^2 is reasonable; exponential 2^n is unreasonable. A heuristic may offer a useful solution sooner without guaranteeing the optimum.

1 mark: Gives 64 for n^2 at $n=8$.

1 mark: Gives 256 for 2^n at $n=8$.

1 mark: Correctly classifies n^2 as reasonable and 2^n as unreasonable.

1 mark: Explains useful practical results without a guaranteed optimum as a reason to consider a heuristic.

Q10. 4 marks

Answer: Decidable: some algorithm terminates with the correct yes/no result for every input. Undecidable: no such universal algorithm exists. A slow exact algorithm can still correctly decide every input eventually. Some instances of an undecidable problem can be solved even though no algorithm solves them all.

1 mark: Defines decidability using a correct terminating answer for every input.

1 mark: Defines undecidability as no algorithm with that universal guarantee.

1 mark: Explains that high runtime alone does not remove decidability.

1 mark: States that some instances may be solvable without an algorithm for all instances.

Q11. 4 marks

Answer: PROCEDURE countGoals(sessions, goal)

```
{  
  hits ← 0  
  FOR EACH minutes IN sessions  
  {  
    IF(minutes >= goal)  
    {  
      hits ← hits + 1  
    }  
  }  
  RETURN(hits)  
}
```

1 mark: Defines countGoals with sessions and goal parameters and initializes hits to 0 before the traversal.

1 mark: Uses FOR EACH to traverse sessions.

1 mark: Tests the current value $\geq$ goal and adds 1 to hits only when true.

1 mark: Returns hits after the complete traversal, with correct block structure.

Q12. 4 marks

Answer: (a) 2, On track. (b) 0, Try again. (c) One list and traversal handle the collection without separate variables and repeated statements for each session. (d) The caller can change the threshold while reusing the same procedure logic.

1 mark: Gives both count 2 and On track for the boundary test.

1 mark: Gives both count 0 and Try again for the zero-hit test.

1 mark: Explains avoiding separate per-session variables or duplicated processing through a list and traversal.

1 mark: Explains reuse of unchanged procedure steps with different goal arguments.